UNIVERSITY OF ABERDEEN **ACADEMIC YEAR 2024–25**

EX3502

Degree Examination in EX3502 Separation Processes 1

9th May 2025 **Time: 14:00 – 17:00**

PLEASE NOTE THE FOLLOWING

(i) You **must not** have in your possession any material other than that expressly permitted in the rules appropriate to this examination. Where this is permitted, such material **must not** be amended, annotated or modified in any way.

(ii) You **must not** have in your possession any material that could be determined as giving you an advantage in the examination.

(iii) You **must not** attempt to communicate with any candidate during the exam, either orally or by passing written material, or by showing material to another candidate, nor must you attempt to view another candidate's work.

(iv) You **must not** take to your examination desk any electronic devices such as mobile phones or other "smart" devices. The only exception to this rule is an approved calculator.

FURTHER REQUIREMENTS SPECIFIC TO THE SCHOOL OF ENGINEERING

a) Candidates **ARE** *permitted to use only an approved calculator.*

b) Candidates **ARE NOT** *permitted to use the Engineering Mathematics Examinations Handbook.*

c) Candidates **ARE NOT** *permitted to use GREEN or RED pen in their exam booklet.*

d) Data sheets are attached to the paper.

e) All question papers must be submitted with the exam booklet.

Candidates must attempt *ALL* questions.

Question 1
Evaporator
A mixture of water and benzoic acid needs to be concentrated from x_F = 15.0 wt% to x_L = 80.0 wt%. A single stage evaporator is proposed to achieve this as benzoic acid can be approximated as a non-volatile solute in water. The feed temperature is 20 °C and preliminary calculations suggest the most economical evaporator operating pressure is 0.2 bar absolute. You can assume all streams have the thermodynamic properties of water/steam and use the attached steam tables in the datasheet where required.

a) A liquid flowrate of L = 394.0 kg hr^{-1} is desired, what is the required feed flowrate F and vapour flowrate V in kg hr^{-1}? **[5 marks]**

b) Calculate the duty of the evaporator in kW. You must assume the Boiling Point Rise (BPR) is given by the following equation:

$$\text{BPR}(^\circ\text{C}) = 15.2x^2 + 8.2x \qquad (1)$$

where x is the weight fraction of "solids" in the stream. **[12 marks]**

c) If saturated steam is available at 30 bara to power the process, what is the steam economy? **[5 marks]**

[Question total: 22 marks]

Question 2
Absorber
Cream from a dairy must have a odourous taint removed using steam. The cream has a concentration of 0.1 mol% and this must be reduced to below 0.018 mol% to prevent consumer complaints. A stripping column that can process 200.0 mol/s of cream is needed, and the Henrys constant for the system is 0.28 mol%/mol%.

a) What is the minimum flowrate of steam capable of achieving the desired outlet concentration in mol/s? **[5 marks]**

b) If a vapour flowrate of 4.0 mol/s is used, how many stages are required?The VLE data is given in Fig. 1 (pg. 11) and you should include this in your solution booklet. **[12 marks]**

c) Packing is available for the absorber which has a height equivalent to a theoretical stage of 0.5 m/stage. Calculate the height of packing required. Discuss what additional height/space is required around the bed and how the diameter of the column is designed (i.e. what are the key considerations in selecting the diameter). **[5 marks]**

[Question total: 22 marks]

Question 3
Multi-stage distillation with Murphree efficiencies
A paint factory is purifying a mixture of 40 mol% pentane and 60 mol% hexane. Distillation is to be used and the bottoms product must reach 20 mol% pentane while the top product must be 10 mol% hexane. The feed is preheated so that upon entry equal molar amounts of vapour and liquid are produced. Three VLE diagrams have been provided in Figs. 2, 3, and 4 (pgs. 12, 13, 14) which you should use and include in your solution booklet.

a) What is the minimum reflux ratio required for this separation? You must submit your graphical construction with your solution booklet. **[6 marks]**

b) If the reflux ratio is $R = 3.8$, how many theoretical stages are required to carry out the distillation? You must submit your graphical construction with your solution booklet. **[6 marks]**

c) Assuming a Murphree tray efficiency of 50% what is the real number of trays in the column? You should assume a partial reboiler is fitted. You must submit your graphical construction with your solution booklet. **[10 marks]**

[Question total: 22 marks]

Question 4
Ponchon-Savarit Distillation
As part of the disposal of old fire-extinguishers, a mixture of carbon-tetrachloride and toluene requires purification before further treatment. The mixture has a concentration of 50.0 mol% CCl_4 and must be purified to 80.0 mol% in the top product while the bottoms product must reach a purity of 20.0 mol% before further processing. You must design a distillation column for this operation.

A VLE chart has been provided in Fig. 5 (pg. 15), while a *H-x -y* chart is available in Fig. 6 (pg. 16) which you should use and include in your solution booklet.

a) Under what circumstances does the Ponchon-Savarit method provide a better estimate of the number of theoretical stages required for a given distillation application, when compared to the McCabe-Thiele method? **[4 marks]**

b) What is the minimum number of stages required to carry out this separation? You must submit your graphical construction with your solution booklet. **[4 marks]**

c) Assuming that a reflux ratio of 2.4 is used, and the feed has an enthalpy of h_F = 20000kJ $kmol^{-1}$ use the Ponchon-Savarit method to determine how many ideal stages are required to carry out the separation. You must submit your graphical construction with your solution booklet **[10 marks]**

d) The column has an overall tray efficiency of 80%. Assuming the column has a thermosyphon (total) reboiler, what is the real number of trays in the column? If you could not solve the previous question please use an estimate of 5 ideal stages.
[4 marks]

[Question total: 22 marks]

END OF PAPER

DATASHEET

Conversion from Celsius to Fahrenheit:

$$^{\circ}F = {}^{\circ}C \times 1.8 + 32$$

Operating lines:

$$y_n = x_{n+1}\frac{R}{R+1} + \frac{x_D}{R+1} \qquad \text{Enrichment line} \qquad (2)$$

$$y_m = x_{m+1}\frac{L_m}{V_m} - x_W\frac{W}{V_m} \qquad \text{Stripping line} \qquad (3)$$

$$y = x\frac{q}{q-1} - \frac{x_F}{q-1} \qquad q\text{-line} \qquad (4)$$

$$\frac{y_{A,n+1}}{1-y_{A,n+1}} = \frac{L'}{V'}\frac{x_{A,n}}{1-x_{A,n}} + \frac{y_{A,1}}{1-y_{A,1}} - \frac{L'}{V'}\frac{x_{A,0}}{1-x_{A,0}} \qquad \text{Absorption} \qquad (5)$$

Relative volatility

$$y_A = \frac{\alpha x_A}{1+(\alpha-1)x_A} \qquad (6)$$

Rayleigh's equation

$$\ln\frac{L_{final}}{L_{initial}} = \int_{x_{initial}}^{x_{final}} \frac{dx}{y-x} \qquad (7)$$

If the relative volatility is constant:

$$\ln\frac{L_{final}}{L_{initial}} = (\alpha-1)^{-1}\ln\frac{x_{final}(1-x_{initial})}{x_{initial}(1-x_{final})} + \ln\frac{1-x_{initial}}{1-x_{final}} \qquad (8)$$

Quadratic equation:

$$a x^2 + b x + c = 0 \qquad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad (9)$$

Ponchon-Savarit equations:

$$P_C = (R+1)(h_V(x=x_D) - h_L(x=x_D)) + h_L(x=x_D) \qquad (10)$$

Table 1: Thermodynamic properties of saturated steam by temperature, calculated using the NASA CEA database and the vapour pressure data of Wexler or Wagner and Pruss (1990). The reference state is the triple point of saturated liquid water.

T (°C)	P (bar)	$C_{p,l}$ (kJ kg^{-1} K^{-1})	$C_{p,v}$ (kJ kg^{-1} K^{-1})	h_l (kJ kg^{-1})	h_{lv} (kJ kg^{-1})	h_v (kJ kg^{-1})	s_l (kJ kg^{-1} K^{-1})	s_v (kJ kg^{-1} K^{-1})
0.01	0.00612	4.22	1.888	0.000611	82501.0	2501.0	-6.161e-08	9.155
1	0.00657	4.216	1.889	4.177	2499.0	2503.0	0.01526	9.129
2	0.00706	4.213	1.89	8.392	2496.0	2505.0	0.03061	9.103
3	0.00758	4.21	1.89	12.6	2494.0	2506.0	0.04589	9.076
4	0.00814	4.208	1.891	16.81	2491.0	2508.0	0.0611	9.051
5	0.00873	4.205	1.892	21.02	2489.0	2510.0	0.07625	9.025
6	0.00935	4.203	1.892	25.22	2487.0	2512.0	0.09134	8.999
7	0.01	4.201	1.893	29.43	2484.0	2514.0	0.1064	8.974
8	0.0107	4.199	1.894	33.63	2482.0	2516.0	0.1213	8.949
9	0.0115	4.197	1.895	37.82	2480.0	2517.0	0.1362	8.924
10	0.0123	4.196	1.896	42.02	2477.0	2519.0	0.1511	8.9
12	0.014	4.193	1.898	50.41	2472.0	2523.0	0.1806	8.851
14	0.016	4.191	1.899	58.79	2468.0	2527.0	0.2099	8.804
16	0.0182	4.188	1.901	67.17	2463.0	2530.0	0.239	8.757
18	0.0206	4.187	1.904	75.55	2458.0	2534.0	0.2678	8.711
20	0.0234	4.185	1.906	83.92	2454.0	2537.0	0.2965	8.666
25	0.0317	4.182	1.912	104.8	2442.0	2547.0	0.3673	8.557
30	0.0425	4.18	1.918	125.7	2430.0	2556.0	0.4368	8.452
35	0.0563	4.179	1.925	146.6	2418.0	2565.0	0.5052	8.352
40	0.0738	4.179	1.932	167.5	2406.0	2574.0	0.5724	8.256
45	0.0959	4.179	1.94	188.4	2394.0	2582.0	0.6386	8.163
50	0.124	4.18	1.948	209.3	2382.0	2591.0	0.7038	8.075
55	0.158	4.181	1.957	230.2	2370.0	2600.0	0.768	7.99
60	0.199	4.183	1.966	251.2	2358.0	2609.0	0.8312	7.908
65	0.25	4.185	1.976	272.1	2345.0	2618.0	0.8935	7.83
70	0.312	4.188	1.987	293.0	2333.0	2626.0	0.955	7.754
75	0.386	4.192	1.999	314.0	2321.0	2635.0	1.016	7.681
80	0.474	4.196	2.012	334.9	2308.0	2643.0	1.075	7.611
85	0.579	4.2	2.026	355.9	2295.0	2651.0	1.134	7.543
90	0.702	4.205	2.042	377.0	2283.0	2660.0	1.193	7.478
95	0.846	4.211	2.059	398.0	2270.0	2668.0	1.25	7.415
100	1.01	4.217	2.077	419.1	2256.0	2676.0	1.307	7.354
110	1.43	4.23	2.121	461.4	2230.0	2691.0	1.419	7.238
120	1.99	4.246	2.174	503.8	2202.0	2706.0	1.528	7.129
130	2.7	4.265	2.237	546.4	2174.0	2720.0	1.635	7.026
140	3.62	4.286	2.311	589.2	2144.0	2733.0	1.739	6.929

Table 2 continued: Thermodynamic properties of saturated steam by temperature.

T (°C)	P (bar)	$C_{p,l}$ (kJ kg^{-1} K^{-1})	$C_{p,v}$ (kJ kg^{-1} K^{-1})	h_l (kJ kg^{-1})	h_{lv} (kJ kg^{-1})	h_v (kJ kg^{-1})	s_l (kJ kg^{-1} K^{-1})	s_v (kJ kg^{-1} K^{-1})
150	4.76	4.31	2.396	632.3	2114.0	2746.0	1.842	6.837
160	6.18	4.338	2.492	675.6	2082.0	2757.0	1.943	6.749
170	7.92	4.369	2.599	719.2	2049.0	2768.0	2.042	6.665
180	10.0	4.406	2.716	763.2	2014.0	2777.0	2.14	6.584
190	12.6	4.447	2.846	807.6	1978.0	2785.0	2.236	6.506
200	15.5	4.494	2.99	852.4	1940.0	2792.0	2.331	6.43
250	39.8	4.865	4.012	1086.0	1715.0	2801.0	2.793	6.072
300	85.9	5.752	6.223	1345.0	1405.0	2750.0	3.255	5.706

Table 2: Thermodynamic properties of saturated steam by pressure, calculated using the NASA CEA database and the vapour pressure data of Wexler or Wagner and Pruss (1990). The reference state is the triple point of saturated liquid water.

P (bar)	T (°C)	$C_{p,l}$ (kJ kg^{-1} K^{-1})	$C_{p,v}$ (kJ kg^{-1} K^{-1})	h_l (kJ kg^{-1})	h_{lv} (kJ kg^{-1})	h_v (kJ kg^{-1})	s_l (kJ kg^{-1} K^{-1})	s_v (kJ kg^{-1} K^{-1})
0.01	6.97	4.201	1.893	29.3	2484.0	2514.0	0.1059	8.975
0.015	13.0	4.192	1.898	54.69	2470.0	2525.0	0.1956	8.827
0.02	17.5	4.187	1.903	73.43	2459.0	2533.0	0.2606	8.723
0.025	21.1	4.184	1.907	88.43	2451.0	2539.0	0.3119	8.642
0.03	24.1	4.183	1.91	101.0	2444.0	2545.0	0.3543	8.577
0.035	26.7	4.181	1.914	111.8	2438.0	2550.0	0.3907	8.521
0.04	29.0	4.181	1.917	121.4	2432.0	2554.0	0.4224	8.473
0.045	31.0	4.18	1.919	130.0	2427.0	2557.0	0.4507	8.431
0.05	32.9	4.18	1.922	137.8	2423.0	2561.0	0.4763	8.394
0.055	34.6	4.179	1.924	144.9	2419.0	2564.0	0.4995	8.36
0.06	36.2	4.179	1.927	151.5	2415.0	2567.0	0.5209	8.329
0.065	37.6	4.179	1.929	157.6	2412.0	2569.0	0.5407	8.301
0.07	39.0	4.179	1.931	163.4	2408.0	2572.0	0.5591	8.275
0.075	40.3	4.179	1.933	168.8	2405.0	2574.0	0.5763	8.25
0.08	41.5	4.179	1.935	173.9	2402.0	2576.0	0.5925	8.227
0.085	42.7	4.179	1.936	178.7	2400.0	2578.0	0.6078	8.206
0.09	43.8	4.179	1.938	183.3	2397.0	2580.0	0.6223	8.186
0.095	44.8	4.179	1.94	187.6	2394.0	2582.0	0.6361	8.167
0.12	49.4	4.18	1.947	206.9	2383.0	2590.0	0.6963	8.085
0.14	52.5	4.18	1.953	220.0	2376.0	2596.0	0.7366	8.031
0.16	55.3	4.181	1.958	231.6	2369.0	2601.0	0.772	7.985
0.18	57.8	4.182	1.962	241.9	2363.0	2605.0	0.8035	7.944
0.2	60.1	4.183	1.966	251.4	2358.0	2609.0	0.832	7.907
0.22	62.1	4.184	1.971	260.1	2352.0	2613.0	0.8579	7.874

Table 2 continued: Thermodynamic properties of saturated steam by pressure.

P (bar)	T (°C)	$C_{p,l}$ (kJ kg^{-1} K^{-1})	$C_{p,v}$ (kJ kg^{-1} K^{-1})	h_l (kJ kg^{-1})	h_{lv} (kJ kg^{-1})	h_v (kJ kg^{-1})	s_l (kJ kg^{-1} K^{-1})	s_v (kJ kg^{-1} K^{-1})
0.24	64.1	4.185	1.974	268.1	2348.0	2616.0	0.8818	7.844
0.26	65.8	4.186	1.978	275.6	2343.0	2619.0	0.904	7.817
0.28	67.5	4.187	1.982	282.6	2339.0	2622.0	0.9246	7.791
0.3	69.1	4.188	1.985	289.2	2335.0	2625.0	0.9439	7.767
0.32	70.6	4.189	1.989	295.5	2332.0	2627.0	0.9621	7.745
0.34	72.0	4.19	1.992	301.4	2328.0	2630.0	0.9793	7.725
0.36	73.3	4.19	1.995	307.0	2325.0	2632.0	0.9956	7.705
0.38	74.6	4.191	1.998	312.4	2322.0	2634.0	1.011	7.686
0.4	75.9	4.192	2.001	317.6	2318.0	2636.0	1.026	7.669
0.42	77.0	4.193	2.004	322.5	2316.0	2638.0	1.04	7.652
0.44	78.2	4.194	2.007	327.2	2313.0	2640.0	1.054	7.636
0.46	79.3	4.195	2.01	331.8	2310.0	2642.0	1.067	7.621
0.48	80.3	4.196	2.013	336.2	2307.0	2644.0	1.079	7.607
0.5	81.3	4.197	2.016	340.5	2305.0	2645.0	1.091	7.593
0.55	83.7	4.199	2.022	350.5	2299.0	2649.0	1.119	7.561
0.6	85.9	4.201	2.029	359.8	2293.0	2653.0	1.145	7.531
0.65	88.0	4.203	2.035	368.5	2288.0	2656.0	1.169	7.504
0.7	89.9	4.205	2.041	376.7	2283.0	2659.0	1.192	7.479
0.75	91.8	4.207	2.047	384.4	2278.0	2662.0	1.213	7.456
0.8	93.5	4.209	2.053	391.6	2274.0	2665.0	1.233	7.434
0.85	95.1	4.211	2.059	398.5	2269.0	2668.0	1.252	7.413
0.9	96.7	4.213	2.065	405.1	2265.0	2670.0	1.269	7.394
0.95	98.2	4.214	2.07	411.4	2261.0	2673.0	1.286	7.376
1	99.6	4.216	2.076	417.4	2258.0	2675.0	1.303	7.359
1.1	102.0	4.22	2.087	428.8	2250.0	2679.0	1.333	7.327
1.2	105.0	4.223	2.097	439.3	2244.0	2683.0	1.361	7.298
1.3	107.0	4.226	2.108	449.1	2238.0	2687.0	1.387	7.271
1.4	109.0	4.229	2.118	458.4	2232.0	2690.0	1.411	7.246
1.5	111.0	4.232	2.128	467.1	2226.0	2693.0	1.434	7.223
1.6	113.0	4.235	2.138	475.3	2221.0	2696.0	1.455	7.201
1.7	115.0	4.238	2.147	483.2	2216.0	2699.0	1.475	7.181
1.8	117.0	4.241	2.157	490.7	2211.0	2701.0	1.494	7.162
1.9	119.0	4.244	2.166	497.8	2206.0	2704.0	1.513	7.144
2	120.0	4.247	2.175	504.7	2202.0	2706.0	1.53	7.127
2.5	127.0	4.26	2.22	535.4	2181.0	2717.0	1.607	7.052
3	134.0	4.272	2.262	561.5	2163.0	2725.0	1.672	6.992
3.5	139.0	4.283	2.302	584.3	2148.0	2732.0	1.727	6.94
4	144.0	4.294	2.34	604.7	2133.0	2738.0	1.777	6.895
4.5	148.0	4.305	2.377	623.2	2120.0	2743.0	1.821	6.856
5	152.0	4.315	2.413	640.2	2108.0	2748.0	1.861	6.821

Table 2 continued: Thermodynamic properties of saturated steam by pressure.

P (bar)	T (°C)	$C_{p,l}$ (kJ kg^{-1} K^{-1})	$C_{p,v}$ (kJ kg^{-1} K^{-1})	h_l (kJ kg^{-1})	h_{lv} (kJ kg^{-1})	h_v (kJ kg^{-1})	s_l (kJ kg^{-1} K^{-1})	s_v (kJ kg^{-1} K^{-1})
6	159.0	4.335	2.48	670.5	2086.0	2756.0	1.931	6.759
7	165.0	4.353	2.543	697.1	2066.0	2763.0	1.992	6.707
8	170.0	4.371	2.603	721.0	2047.0	2768.0	2.046	6.662
9	175.0	4.388	2.66	742.7	2030.0	2773.0	2.094	6.621
10	180.0	4.405	2.715	762.7	2014.0	2777.0	2.138	6.585
15	198.0	4.485	2.964	844.7	1946.0	2791.0	2.315	6.443
20	212.0	4.562	3.19	908.6	1890.0	2798.0	2.447	6.339
25	224.0	4.638	3.404	962.0	1840.0	2802.0	2.554	6.256
30	234.0	4.714	3.612	1008.0	1795.0	2803.0	2.646	6.186
35	243.0	4.791	3.817	1050.0	1753.0	2803.0	2.725	6.125
40	250.0	4.869	4.022	1087.0	1713.0	2801.0	2.797	6.07
45	257.0	4.949	4.228	1122.0	1676.0	2798.0	2.861	6.02
50	264.0	5.032	4.438	1155.0	1640.0	2794.0	2.921	5.974
60	276.0	5.208	4.877	1214.0	1571.0	2785.0	3.027	5.89
70	286.0	5.4	5.354	1267.0	1505.0	2773.0	3.122	5.815
80	295.0	5.614	5.883	1317.0	1442.0	2759.0	3.208	5.745
90	303.0	5.854	6.476	1364.0	1379.0	2743.0	3.287	5.679
100	311.0	6.127	7.147	1408.0	1318.0	2725.0	3.36	5.616
120	325.0	6.813	8.819	1491.0	1194.0	2686.0	3.496	5.494

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

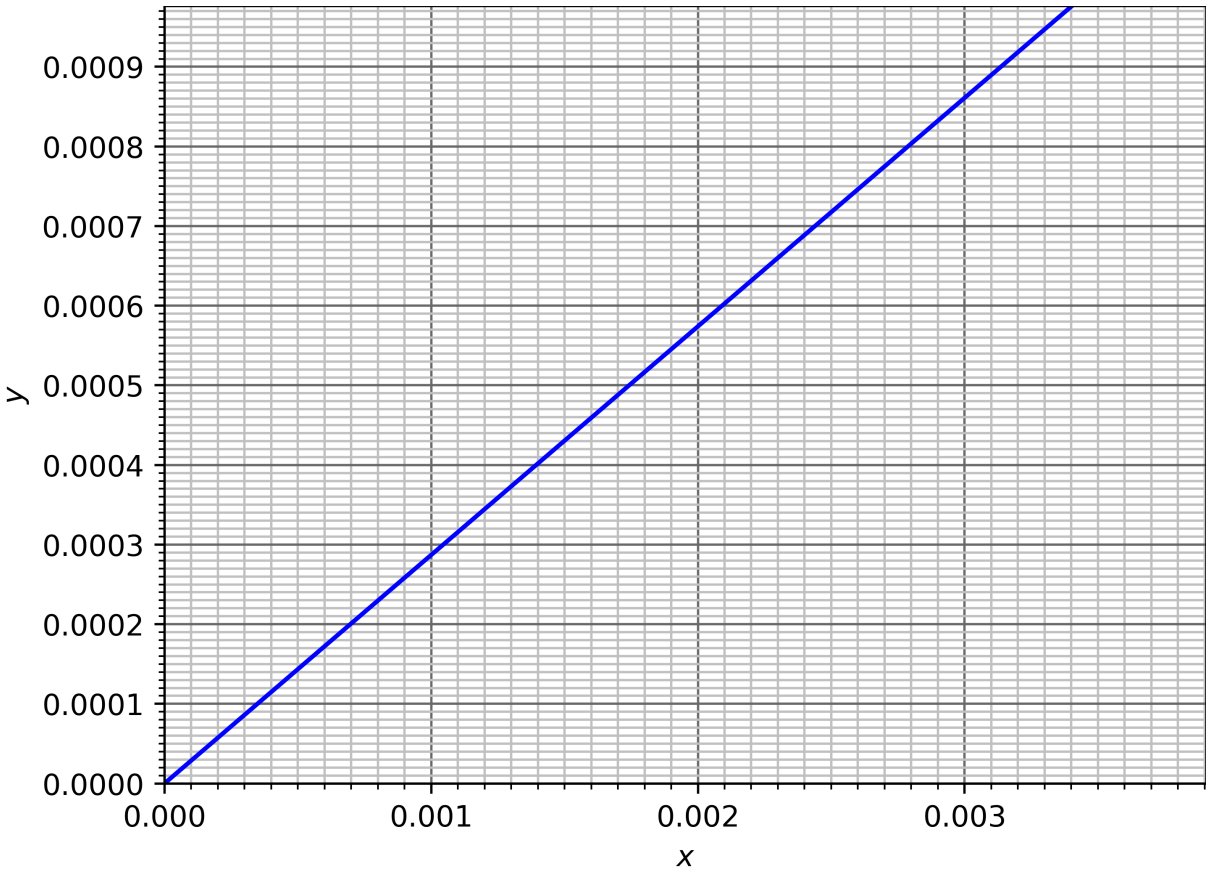

Figure 1: A blank VLE chart for use in Q. 2

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

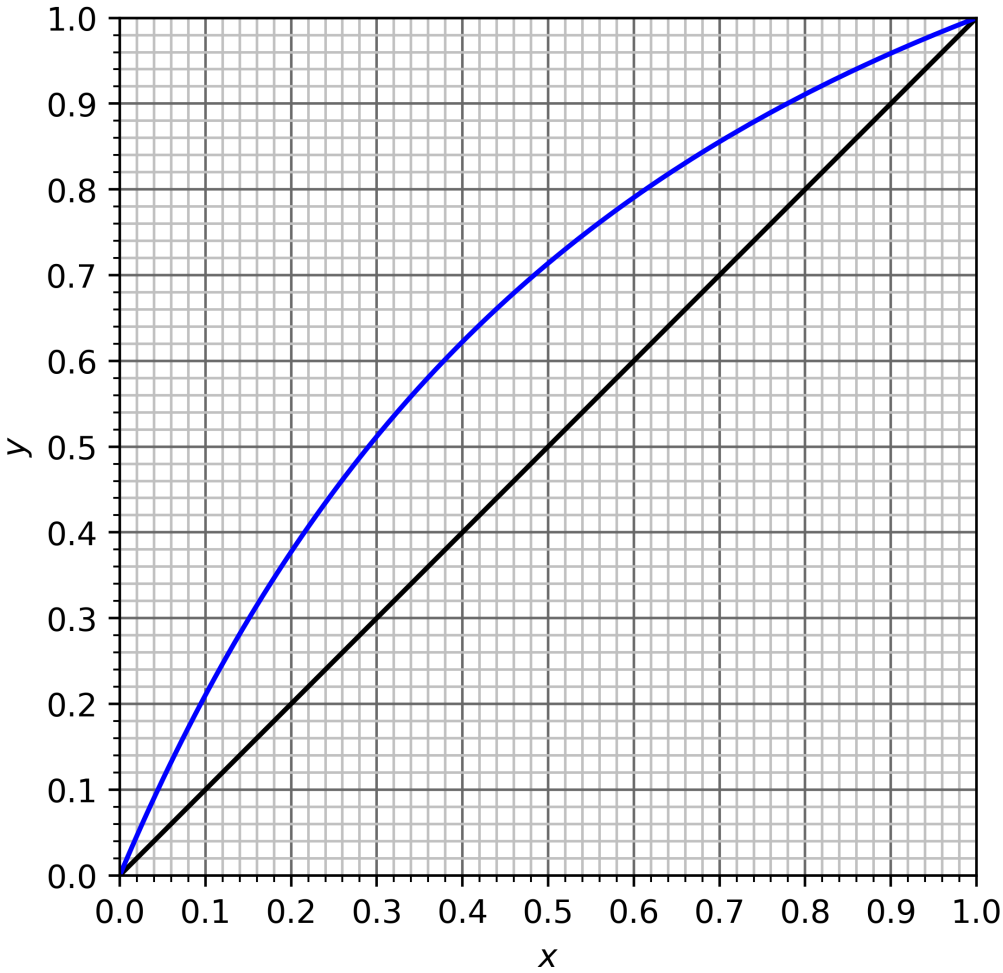

Figure 2: VLE data for the benzene-toluene system. For use in Q. 3

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

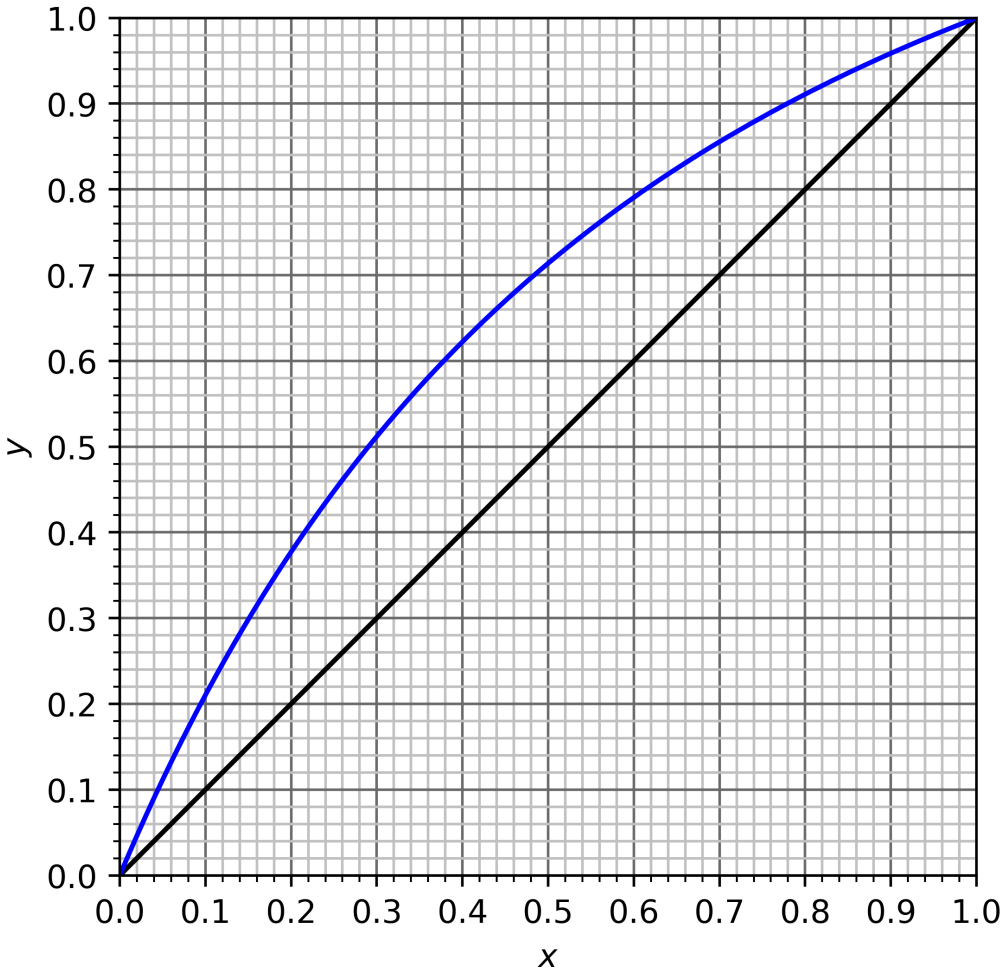

Figure 3: VLE data for the benzene-toluene system. For use in Q. 3

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

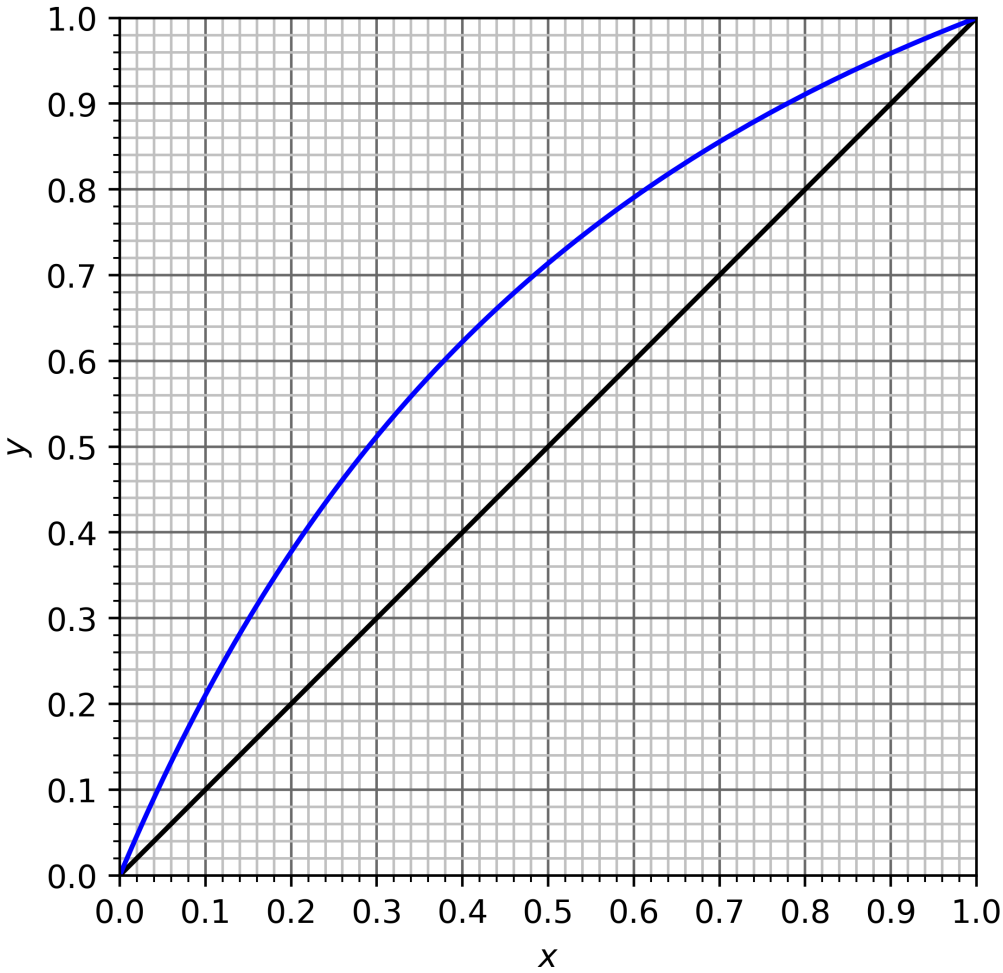

Figure 4: VLE data for the benzene-toluene system. For use in Q. 3

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

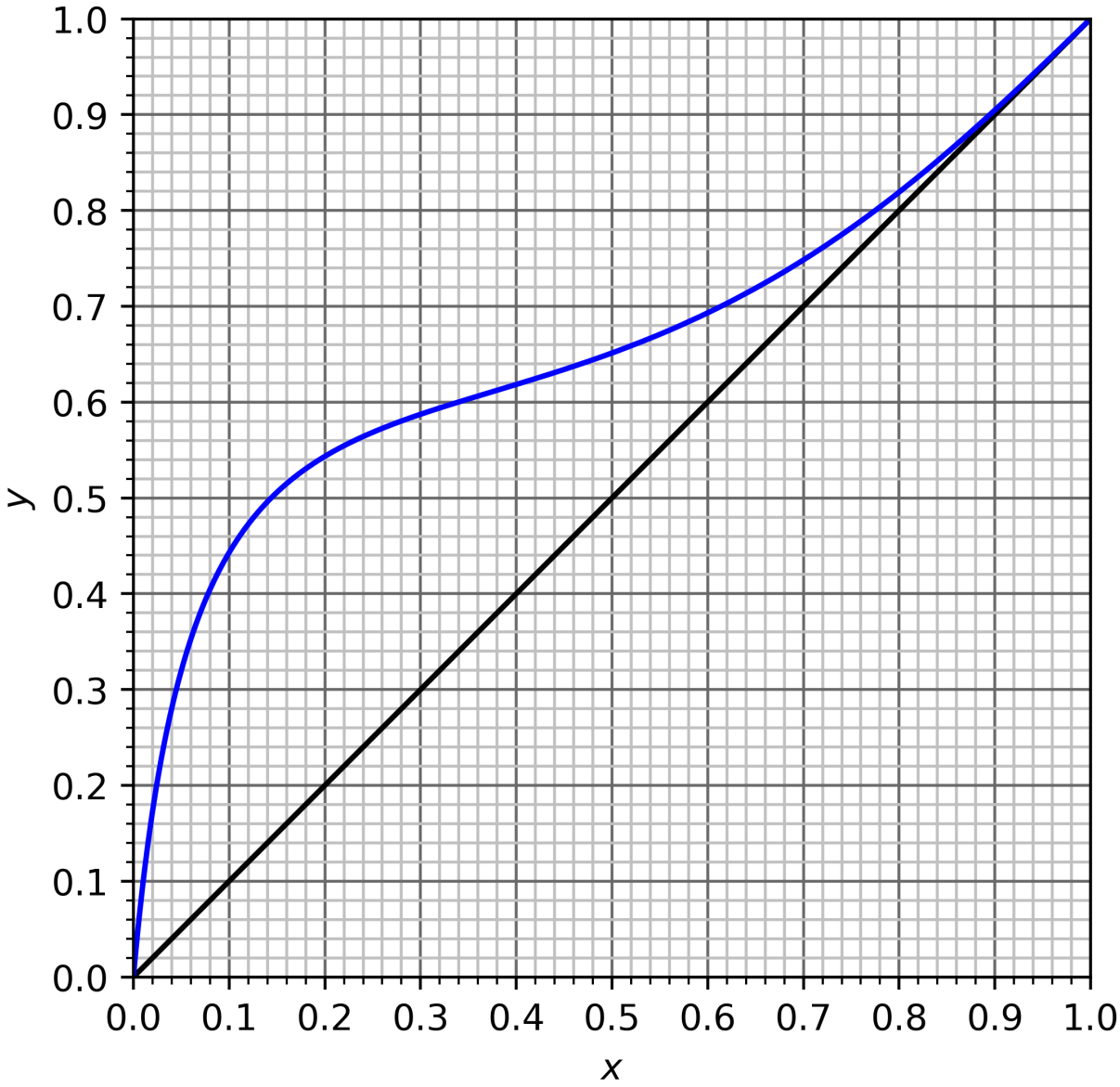

Figure 5: VLE data for the carbon-tetrachloride/toluene system. For use in Q. 4

Data sheet handout.
Submit this graph as part of your solution book.

Student ID:

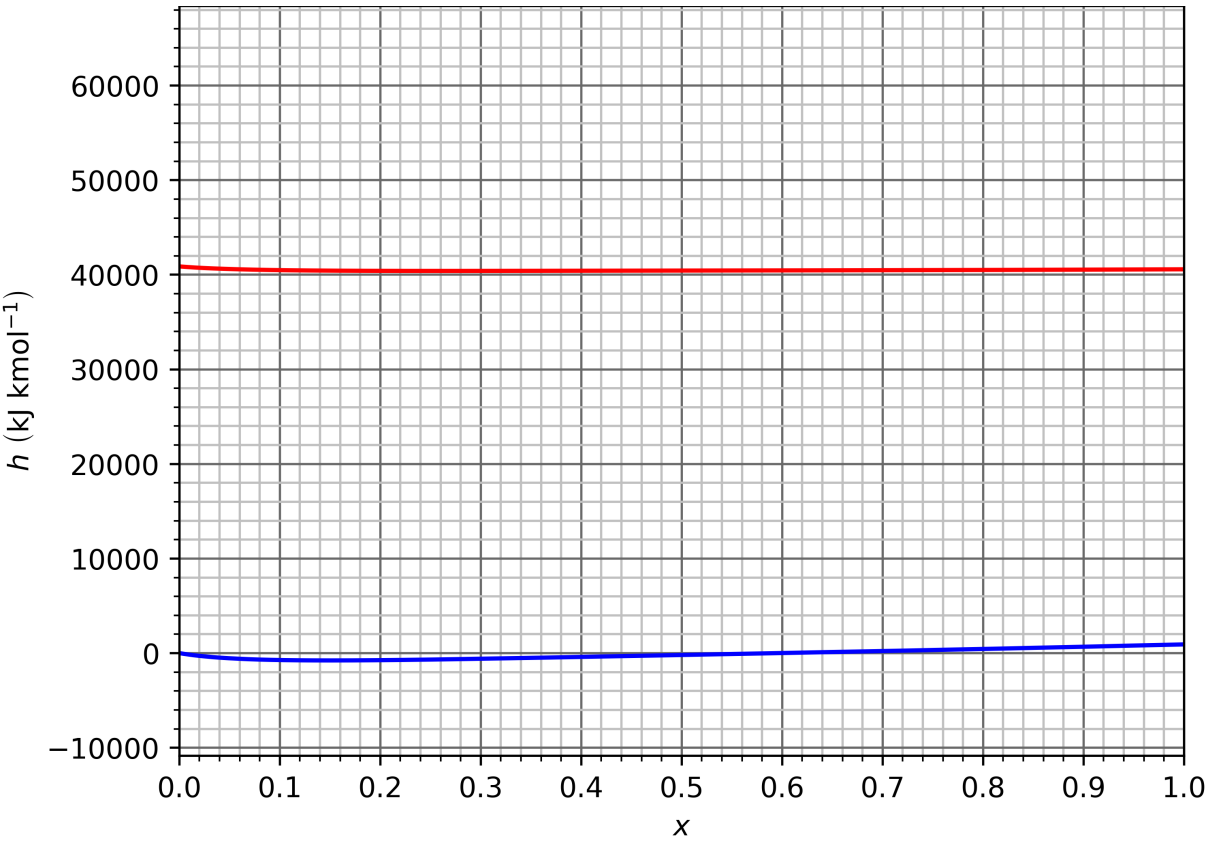

Figure 6: *H-x -y* data for the carbon-tetrachloride/toluene water system. For use in Q. 4